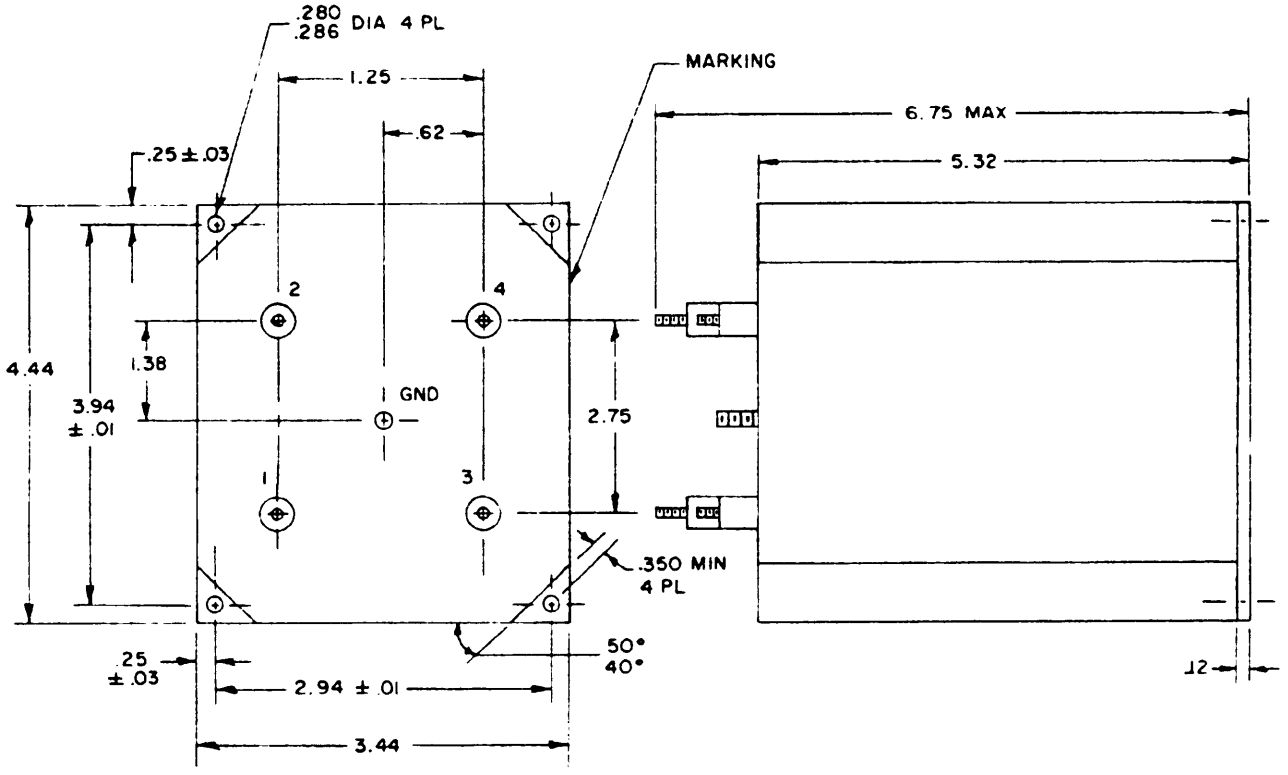
MILITARY SPECIFICATION SHEET

TRANSFORMERS AND INDUCTORS
(AUDIO, POWER AND HIGH-POWER PULSE)
INDUCTOR, CHARGING, TF4S37YY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the inductor described herein shall consist of this document and the latest issue of Specification MIL-T-27.

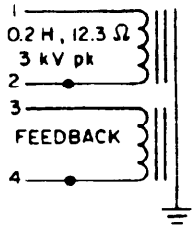


NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Marking shall be on the side of the case.
4. Unless otherwise specified, tolerance on all dimensions shall be ±.06 inch (1.5 mm).
5. Terminals 1, 2: 801-B0666-5, ceramaseal or equal.
6. Terminals 3, 4: 801-B0666-8, ceramaseal or equal.
7. GND stud: .190-32 UNF-2A x .50 (12.7 mm) cres.
8. GND stud on C_L ±.03 (.8 mm).

FIGURE 1. Dimensions and configuration.

FSC 5950

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Working voltage (peak):

(1-GND): 3000 V
(2-GND): 6000 V
(4-GND): 600 V

CIRCUIT DIAGRAM AND MARKING

INCHES	MM	INCHES	MM
.01	.3	1.25	31.8
.03	.8	1.38	35.1
.06	1.5	2.75	69.8
.12	3.0	2.94	74.7
.25	6.4	3.44	87.4
.280	7.11	3.94	100.1
.286	7.26	4.44	112.8
.350	8.89	5.32	135.1
.50	12.7	6.75	171.4
.62	15.7		

FIGURE 1. Dimensions and configuration - Continued.

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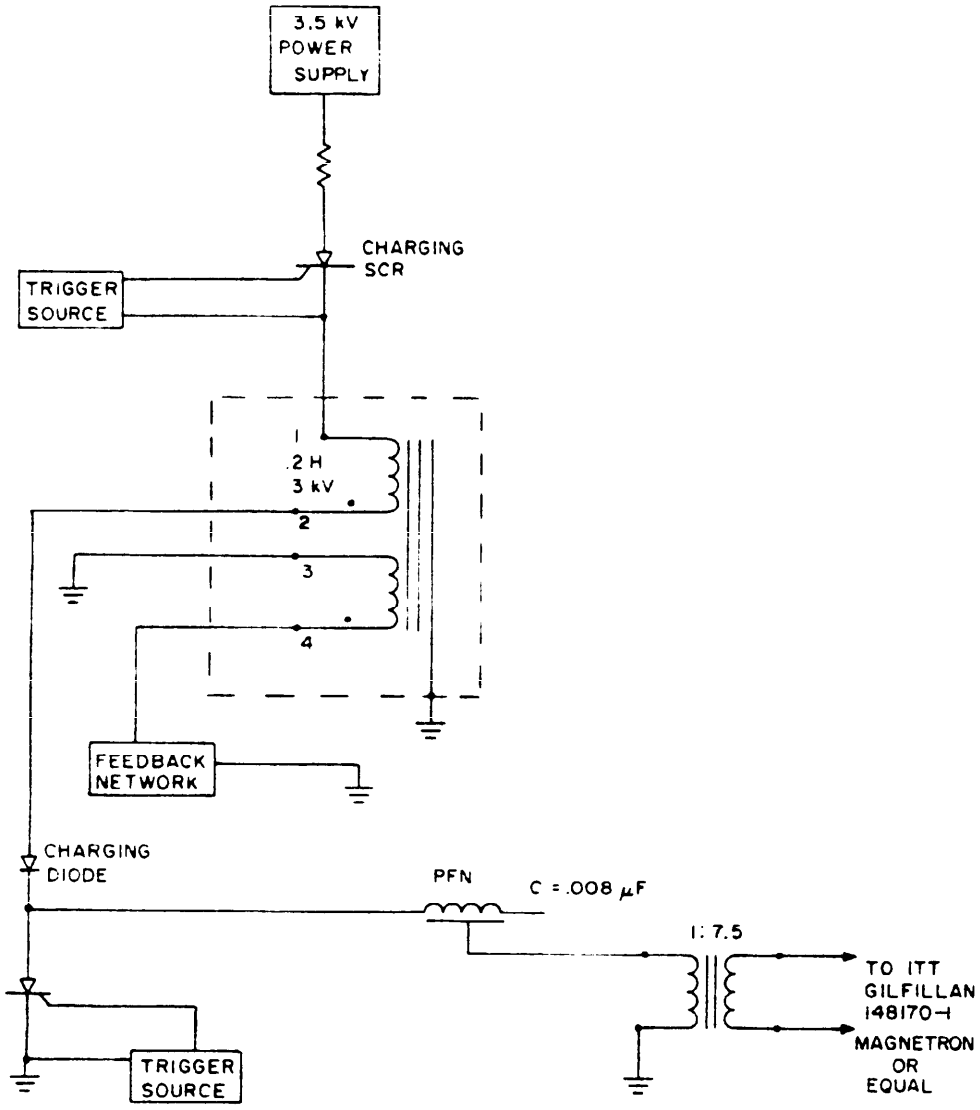


FIGURE 2. Test circuit.

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:
- Inductance (1-2): 0.2 henry ±5 percent.
- Power (1-2): 158 watts maximum.
- Voltage:
  - (1-2): 3 kilovolts peak at 3,300 pulses per second.
  - (3-4): 300 volts peak.
- Current (1-2): 0.5 ampere peak.
- DC resistance:
  - (1-2): 12.3 ohms maximum.
  - (3-4): 0.12 ohm maximum.
- Working voltage (peak):
  - (1-GND): 3,000 volts.
  - (2-GND): 6,000 volts.
  - (4-GND): 600 volts.
- Frequency range: 1,000 hertz to 40 kilohertz.

Design and construction:
- Dimensions and configurations: See figure 1.
- Duty cycle: Continuous.
- Case: Metal encased.
- Terminals: Screw thread.
  - 1 and 2: Ceramaseal 801-B0666-5 or equivalent.
  - 3 and 4: Ceramaseal 801-B0666-8 or equivalent.
  - Terminal height: 1.430 inches maximum.
- Weight: 20 pounds maximum.
- Operating temperature range: -55° to +130°C.

Terminal strength: MIL-STD-202, method 211, test condition E, 15 pound-inches.

Dielectric withstanding voltage (at sea level):
- (1-2) to (3-4) and GND: 9,400 volts rms.
- (3-4) to GND: 1,680 volts rms.

Induced voltage: With 6,000 volts peak, 88.9 microseconds wide, at 3,300 pulses per second applied to (1-2) for 60 seconds, the induced voltage across (3-4) shall be 600 volts peak, a minimum of 44.4 microseconds wide, at 3,300 pulses per second.

Electrical characteristics:
- Inductance: With 132 volts rms, 260 hertz, and 0.5 ampere peak applied to (1-2), the inductance across (1-2) shall be 0.2 henry ±5 percent with terminals 3 and 4 open.
  - Inductance change (1-2): ±10 percent from 250 milliamperes to 500 milliamperes.
- No load: The voltage ratio shall be 10:1 ±5 percent.
- Frequency response: ±3 dB from 1 kilohertz to 40 kilohertz.
- Storage factor (1-2): The storage factor shall be a minimum of 30 seconds and a maximum of 50 seconds at 1 kilohertz.
- Wave shape: 1/2 sine wave, 125 microseconds wide at 3 kilovolts peak in the test circuit (see figure 2).
- Capacitance: Less than 300 picofarads at 1 kilohertz with terminals 1 and 2 (tied) to ground.
- DC resistance:
  - (1-2): 12.3 ohms, maximum.
  - (3-4): 0.12 ohm, maximum.
- Polarity: Additive with 2 and 3 connected.

Temperature rise: 59°C with 3 kilovolts peak, 3,300 pulses per second, 0.5 ampere peak applied to (1-2) at an ambient temperature of +71°C maximum.

Corona discharge: Applicable, test circuit 2.

Salt spray (corrosion): MIL-STD-202, method 101, test condition B.

Marking location: See figure 1.

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Part number: M27/219-01.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AR, WC, ME
Navy - MC, AS, SH, OS
Air Force - 19

Preparing activity:
Army - EL

Agent:
DLA - ES

(Project 5950-0555-10)

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE MIL-T-27/219 TRANSFORMERS AND INDUCTORS (AUDIO, POWER AND HIGH-POWER PULSE) INDUCTOR, CHARGING TF4S37YY

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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